

Harmonic oscillator coherent state

A coherent state minimizes uncertainty. The ground state is a coherent state. To make the ground state more interesting, parameters r and θ are added to shift $\langle x \rangle$ and $\langle p \rangle$ from zero.

$$\psi_{n,r,\theta}(x,t) = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{\frac{1}{4}} \left(\frac{\hbar}{m\omega} \right)^{n/2} \\ \times \exp \left[\frac{m\omega}{2\hbar} (x - \langle x \rangle)^2 + \frac{i}{\hbar} \langle p \rangle \left(x - \frac{1}{2} \langle x \rangle \right) - i \left(n + \frac{1}{2} \right) \omega t \right] \frac{d^n}{dx^n} \exp \left[-\frac{m\omega}{\hbar} (x - \langle x \rangle)^2 \right]$$

Parameters r and θ are polar coordinates in phase space such that

$$\langle x \rangle = \sqrt{\frac{2\hbar}{m\omega}} r \cos(\omega t + \theta)$$

and

$$\langle p \rangle = -\sqrt{2m\hbar\omega} r \sin(\omega t + \theta)$$

Verify $\psi_{n,r,\theta}(x,t)$ solves the Schrodinger equation

$$i\hbar \frac{d}{dt} \psi(x,t) = \hat{H} \psi(x,t)$$

where

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2}$$

For the ground state $\psi_{0,r,\theta}(x,t)$ verify that

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{2m\omega}}$$

and

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{m\hbar\omega}{2}}$$

Hence $\Delta x \Delta p$ is the minimum allowed by the uncertainty principle.

$$\Delta x \Delta p = \frac{\hbar}{2}$$